

How to secure your instances of SAP HANA Cloud

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Who am I?

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- SAP Mentor since 2025
- Spokeswoman for DSAG user group SAP HANA operations since 2017
- Author of
 - SAP HANA teaching materials for Geospatial Analysis and Text Mining for universities in the SAP University Alliance program
 - Book SAP HANA – Datenbankadministration, SAP Press ISBN 978-3-8362-6850-9



Myths

- Myth #1: SAP patches everything, so I'm safe!
- Myth #2: Passwords are enough
- Myth #3: Network traffic is secure by default
- Myth #4: Everyone can be admin

Myth #1: “SAP patches everything, so I’m safe!”

- ✓ SAP updates OS & DB engine semi-automatically
- ✗ Doesn't protect against misconfigurations or weak apps

Reality Check #1

- Most attacks come through:
 - Weak app security
 - Misconfigured connections
 - Unsafe development practices
 - Incomplete user management



Myth #2: “Passwords are enough.”

- Passwords are:
 - Weak
 - Often shared
 - Easily stolen
- Passwords alone are not security.

Reality Check #2

- Integrate SAP HANA Cloud/Cloud Identity Services with your corporate Identity Provider
- Use MFA everywhere
- Use OAuth, SSO, certificates instead of only passwords
- Use your password store with enforced password guidelines



Myth #3: “Network traffic is already secure.”

- Secure TLS is available in SAP HANA Cloud
- But not enforced everywhere by default
- Open connections = open invitations

Reality Check #3

- Always enforce secure TLS for:
 - Clients
 - Apps
 - APIs
- Restrict connections to what you really need
- Use secure tunnels for hybrid scenarios

<https://community.sap.com/t5/technology-blog-posts-by-members/configure-an-sap-cap-application-to-connect-to-an-sap-hana-cloud-database/ba-p/13890278>



Myth #4: “Roles and privileges? Just give everyone admin.”

- Quick & convenient? Yes.
- Secure? Absolutely not.

Reality Check #4

- Apply principle of least privilege
- Create roles carefully
- Pay attention to segregation of duties
- Monitor and audit user activity



Best Practices Recap

- ✓ Multi-factor authentication & identity management
- ✓ Enforce secure TLS everywhere
- ✓ Use least privilege roles, segregation of duties & proper management of access
- ✓ Continuous logging & auditing – and also take a look at the logs!

Best Practices Recap

Authentication and user management

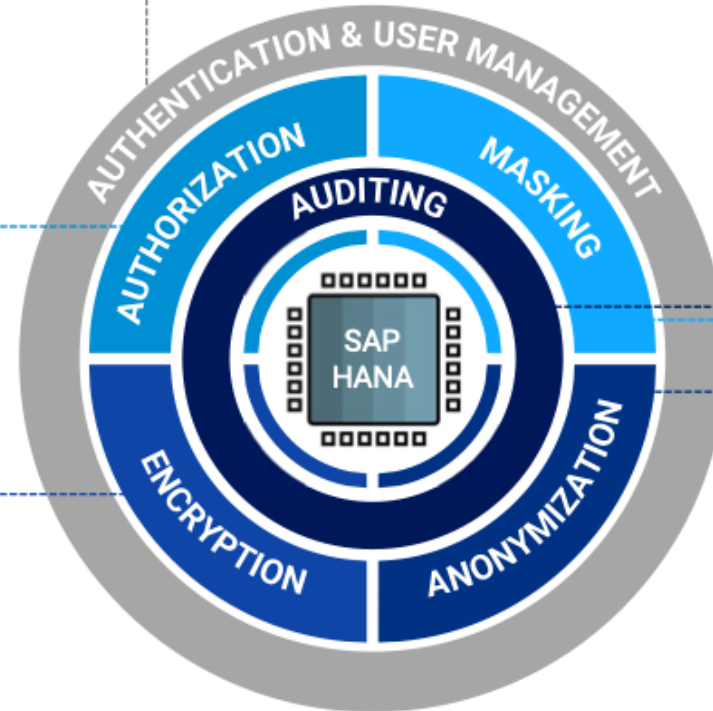
- User and identity management
- Single sign-on (SAML, JWT, X509)
- Identity Access Governance (IAG) and Identity Management (IDM) integration
- LDAP integration
- Password and user connect restriction policies per user group

Authorization

- Role management framework
- Privileges for all user types
- Row-level access control
- Integrated application authorizations
- Authorization troubleshooting

Encryption

- At rest and in motion
- Backup encryption
- Key management
- Communication Encryption



Auditing

- Security logging and analysis for all system events, with customizable policies
- Log read and write access to critical data
- Firefighter logging
- Audit retention policies, audit policy wizard
- Comprehensive logging for cloud operator actions

Masking

- Dynamic data masking for tables and views
- Custom mask expressions

Anonymization

- Real-time data anonymization
- k-anonymity (including l diversity), differential privacy
- Custom definition of anonymization views (calculation and SQL views)
- Full integration with authorization framework
- Reporting
- Data anonymization KPIs

https://assets.dm.ux.sap.com/webinars/sap-user-groups-k4u/pdfs/230615_sap_security_june_examine_the_security_perspectives_of_hana_cloud.pdf



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